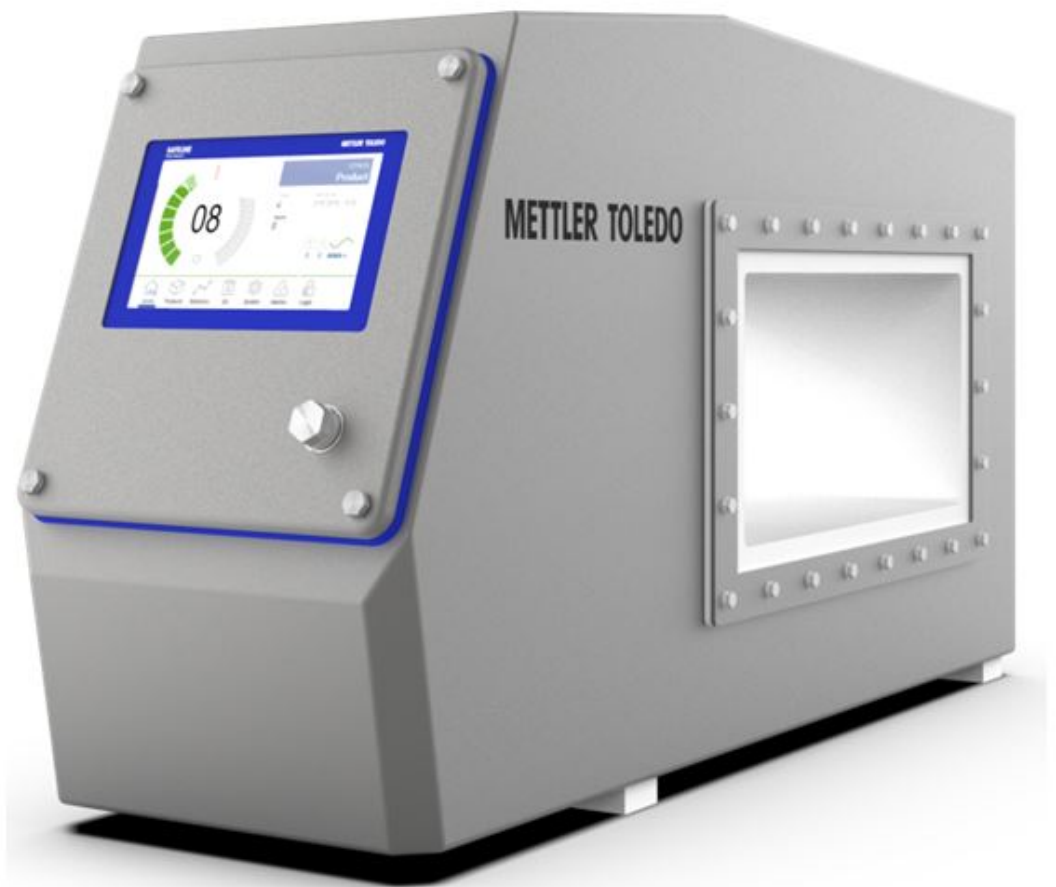


Metal Detector

M31R/M33R/M33RB/M34R



METTLER TOLEDO

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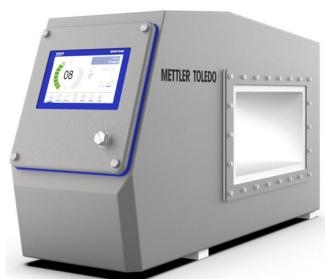
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1 Introduction

This manual covers the following M30 R-series metal detectors:

- M31R
- M33R
- M33RB
- M34R



A M30 R-Series Metal Detector

1.1 Purpose of this Manual

This manual is designed for the M30 R-series of metal detectors. The manual contains important information on how to work with this metal detector effectively and safely. It explains how to prepare for installation, and how to install and use the metal detector. This manual explains how to navigate the pages of the user interface, and basic operation of the metal detector. See the Operation Manual for full instructions for operating the metal detector through the user interface: **www.mt.com/M30-Series-Operation-Manual**

1.2 Caring for the Metal Detector

Contact your authorised representative about the following products and services:

- Installation and commissioning of the metal detector.
- Certified, genuine spare parts.
- Emergency repair and annual service contracts.
- An annual performance verification service to certify that the metal detector has maintained its performance levels.
- Software upgrade packages.

1.3 Contact details

To contact the service representative for your product and your region, go to www.mt.com/contact.

1.4 Mettler-Toledo End User License Agreement (EULA) for Software

The software on this product is subject to the **METTLER TOLEDO** End User License Agreement. Refer to **www.mt.com/legal**

2 Safety Notes

2.1 Safety Labels and Notice Labels

The ISO 3864 safety labels are installed at potentially hazardous areas on the equipment. They give special safety-related notifications. The locations of these labels are given in the drawings supplied with your equipment. There are three types of safety labels:

- Hazard notifications
- Mandatory procedures
- Prohibitive procedures

Additionally, **NOTICE** labels may appear on your equipment.

The meanings of the different kinds of safety labels are explained in the following sections.

Before you transport, install, operate, or work on the equipment, find out about the location and meanings of the safety labels. Maintain the labels so that they are clear of obstructions and are readable. Do not remove any labels. Replace any label that is no longer readable.

2.1.1 Hazard Notifications

A hazard notification consists of the following:

- Hazard alert symbol (yellow triangle with black symbol)
- Signal word (**DANGER**, **WARNING**, or **CAUTION**)
- Special notifications related to the hazard (as required)

The hazard severity panels are attached next to the labels for the hazard alert symbols on the equipment.

2.1.2 Mandatory Procedures

Mandatory procedures alert personnel to special required actions. The labels are round with a blue background and a white symbol describing the required action. The following table lists the mandatory procedures labels which may be installed on your equipment.

	Read the technical manual
---	----------------------------------

2.1.3 Prohibited Procedures

Prohibited procedures alert personnel to particular actions to avoid. The labels are round with a circular red band and single diagonal cross bar. The black symbol describes the action to avoid. The following table lists the prohibited procedure labels that may be installed on your equipment.

	Do not operate the equipment if you have implanted medical devices
---	---

2.2 Definitions of Signal Words

We use signal words and warning symbols for safety notices. Safety notices contain warnings and information about safety issues. Ignoring safety notes can lead to death or personal injury, damage to the instrument, malfunctions and erroneous results.

	(red): This hazard severity panel indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
	(orange): This hazard severity panel indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
	(yellow): This hazard severity panel indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury.
	NOTICE (blue): This word indicates important information that is not related to personal injury which, if ignored, could result in damage to the equipment, damage to property, malfunctions, erroneous results, or loss of data.

2.3 Meaning of Hazard alert Symbols

The following symbols may be installed on your equipment.

Icon	Description	Icon	Description
	No pacemakers, metallic implants or neurostimulators		Read the technical manual
	Non-ionising radiation		Electrical shock

2.4 General Safety Information

General safety information

Always follow the guidelines in this manual. Failure to follow the advice in this manual may lead to the protection offered by the equipment becoming impaired.

There are no user-replaceable serviceable parts. Contact your service representative for any service issues: www.mt.com/slmd-contact.

2.5 Intended Use

2.6 Non-ionizing Radiation



⚠ WARNING

The electromagnetic fields inside and directly outside the aperture of metal detectors can affect the operation of cardiac pacemakers and other active or passive medical devices (implanted or external). Therefore, personnel fitted with these types of devices must always consult their doctor or physician before they operate or come into close proximity of the metal detector aperture.



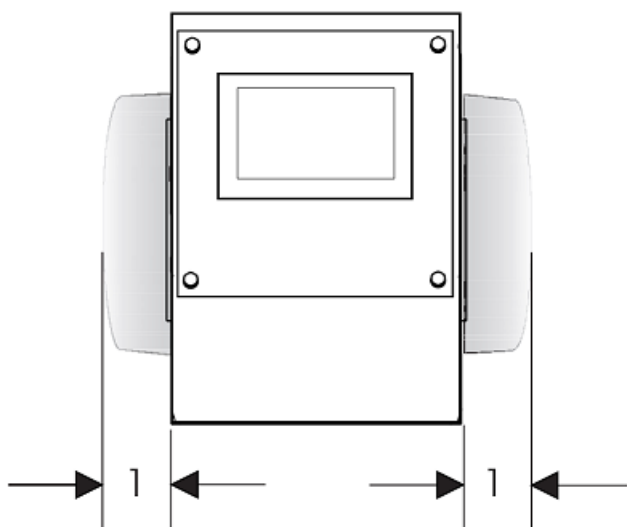
⚠ CAUTION

Non-ionizing radiation.

This is a radio-frequency metal detector that generates electromagnetic fields within the frequency range 10 kHz to 1 MHz. When the metal detector is operating, avoid prolonged exposure to the electromagnetic field inside the aperture and up to 200 mm from the aperture.

Inside the aperture of the detector the magnetic field strength is typically up to 600 A/m (RMS) across all operating frequencies, head types and aperture sizes.

At a distance of 200mm directly outside of the aperture (1) the magnetic field strength is less than 1 A/m (RMS) across all frequencies, head types and apertures.



Non-ionising radiation

The maximum electric field inside the aperture is typically 10 V/m (RMS) with negligible values immediately outside the aperture.

The current and energy induced in the human body resulting from exposure to the field inside the aperture comply with published basic restriction levels for occupational exposure.

If specific values are required per detector type, size and operating frequency then contact **METTLER TOLEDO Safeline**.

2.7 Important Notice

Limitations of use

Using metal detection inspection equipment is an important tool in minimizing the risk of shipping product from your facility containing metal contaminants which could eventually pose a risk to a consumer.

Using this or any other metal detector inspection unit does not guarantee that all contaminated products will be identified and rejected from the product flow. Metal contaminant orientation, geometry and composition will influence the minimum size of contaminant that can and will be detected. In addition, the metal detector settings used to suppress unwanted signals from the product under inspection or the system vibration, may also suppress and in some instances prevent the detection of contaminant signals which are larger than the minimum detectable ball size.

Equipment malfunction and customer responsibility

The unit itself may also experience either continuous or intermittent operational malfunction which will severely impede its ability to detect and reject product containing potentially harmful metal contaminants. To minimize the risk of contaminated product leaving the facility where the metal detector equipment is applied, it is vital that operational procedures are in place to continuously test the performance of the metal detector detection system.

If any test shows a risk of either continuous or intermittent malfunction of the metal detection inspection system procedures to isolate and either discard or re-inspect the product which may contain contaminants must be initiated.

Ensuring that products containing metal contaminants do not leave the facility where the metal detection equipment is installed and applied is therefore outside the reasonable control of **METTLER TOLEDO Safeline**

Minimising the risk of undetected contaminants in your product

METTLER TOLEDO Safeline recommends that use is made of the in-built performance monitoring routine. The feature will help you as a company to comply with quality assurance standards by ensuring that the detector system is tested and operated to your company's specified quality assurance standards and performance. See: Setting up performance tests and Running performance tests.

Customer standard operating procedure requirements

METTLER TOLEDO Safeline recommends that the metal detection system is operated in accordance with a customer defined Standard Operation Procedure (SOP) that, as a minimum, contains the following elements:

- Determine a metal detector test interval – hourly, shift, daily etc.
- Use of certified traceable test samples.
- Records of tests performed and results obtained.
- Records of personnel who are authorized to run tests.
- Procedures for quarantine of "suspect" product and the initiation of corrective and preventive measures if any test fails to meet defined criteria.

Food contact applications

METTLER TOLEDO Safeline declares that this equipment is suitable for use in food-contact applications when operated according to its intended use. Although this equipment was cleaned prior to shipment, **METTLER TOLEDO Safeline** recommends you re-clean it before using it for the first time. For further cleaning instructions, see: [Cleaning ► Page 29]

Your metal detector's annual performance verification

HACCP guidelines require regular checking of product inspection equipment to make sure that the required performance level is being maintained for each critical control point. Our qualified engineers can provide the necessary certification to show that we have checked your metal detector and that we can verify that it meets your compliance needs.

3 Installation

3.1 Lifting and moving the detector



WARNING

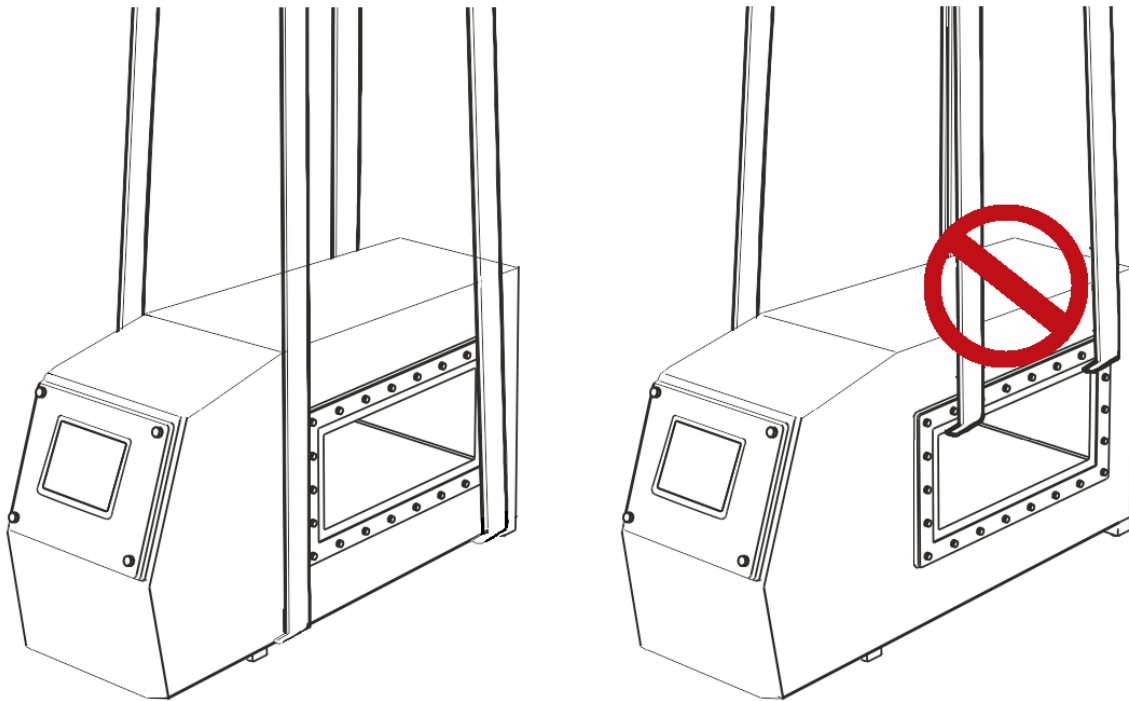
Do not move the metal detector by hand.
The metal detector is heavy. Use suitable lifting equipment.



NOTICE

Do not put lifting straps or supports through the aperture of the metal detector.
Pressure applied to the aperture causes damage to the metal detector.

1. Position the lifting straps under the metal detector, as shown in the following figure.
2. Make sure the metal detector is secure and cannot become unbalanced.



Lifting the metal detector: correct way (left) and incorrect way (right)

3.2 Positioning and Environmental Requirements

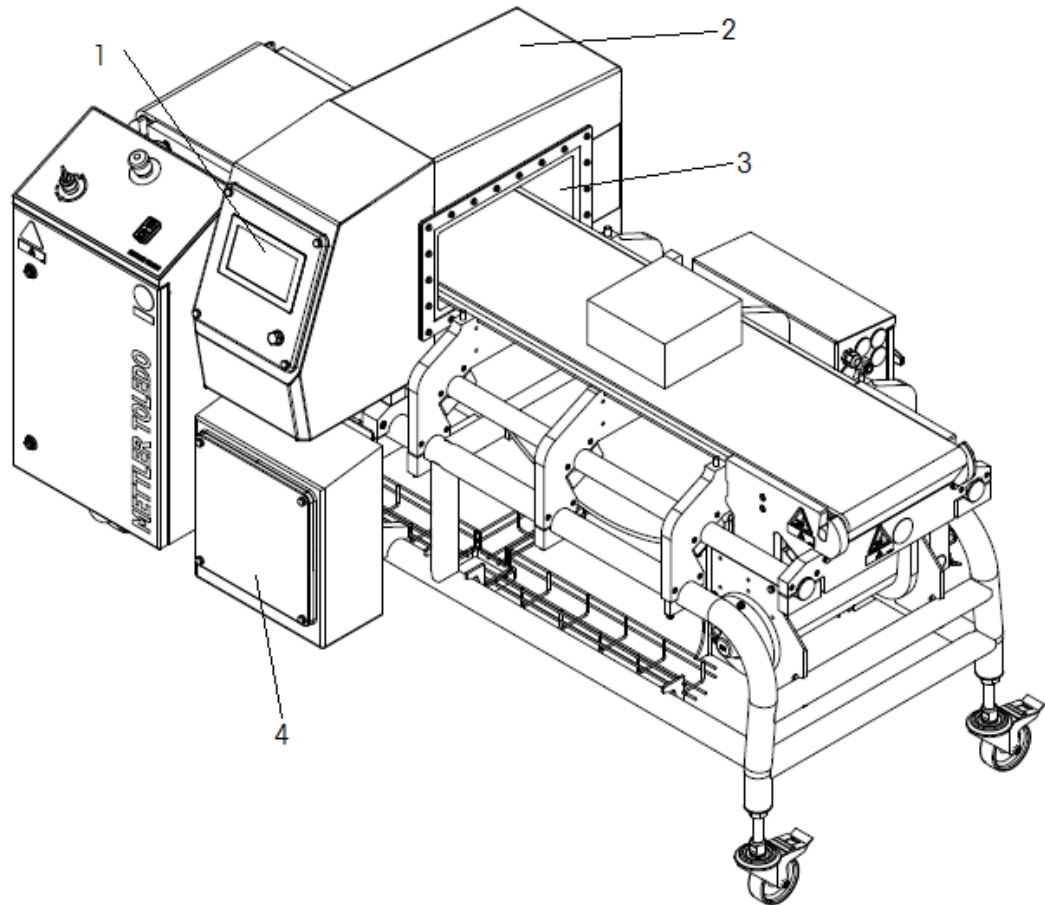


NOTICE

Poor performance of the metal detector can result from vibration and from electromagnetic interference.

- Do not install any parts of the metal detector near to devices which emit electromagnetic interference, particularly in the same frequency range as the metal detector. For example: radio transmitters.
- Take care to follow the manufacturer's instructions when installing inverters and variable-speed drives near the metal detector.

- Where possible, do not place cables from inverters, variable-speed drives and similar devices near the metal detector or its cables.
- Avoid supporting the main metal detector (2) on or near vibrating structures or machines affected by mechanical shock.
- Position the metal detector's user interface (1) so that it is accessible.
- Position the power supply unit enclosure (4) to allow ease of access when servicing.
- Make sure the floor is flat and level.
- Do not run metal detector's cables next to other cables carrying switched or heavy loads.
- Do not run cables near the aperture (3) on the following representative figure.



Metal detector on a METTLER TOLEDO conveyor

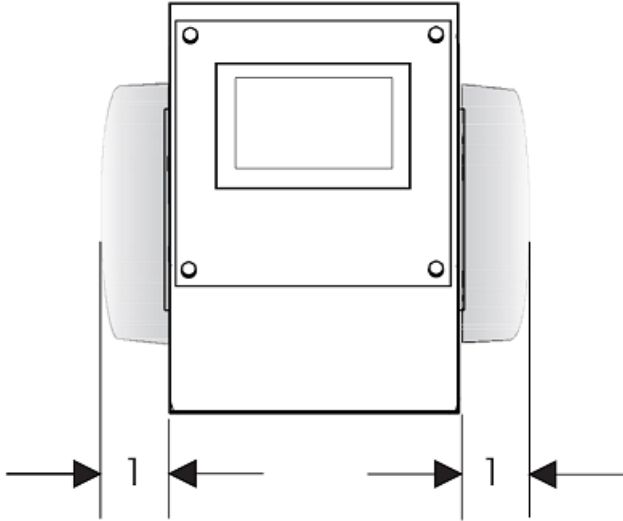
3.2.1 Metal-free zones



NOTICE

For correct performance of the metal detector, keep the area around the aperture of the metal detector free from metal. This includes any metal conduit or electrical cable.

Basic calculations for the metal-free zone, (1) in the following figure:



- Stationary metal: Metal free zone = 1.5 x aperture height/width (the smallest dimension of the two).
- Moving metal: Metal free zone = 2.0 x aperture height/width (the smallest dimension of the two).

Various factors can influence the metal-free calculation, such the aperture height, the type of metal detector, the sensitivity and the phase. Reducing the metal detector's sensitivity, for example, reduces the metal-free zone.

3.2.2 Environmental Requirements



NOTICE

The metal detector is designed to operate correctly within the environmental conditions shown here.

Refer also to the chapter: [Technical Data ► Page 30]

Humidity range

Maximum relative humidity: 93% for temperatures up to 45°C.

Indoor altitude

Up to 2000 m.

Pollution degree

Pollution degree 2 in accordance with IEC 664.

Warm-up time

No warm-up time required at an ambient temperature of 20°C.

Sound output

Less than 62 dBA at a distance of 1 m, unless you install an audible alarm louder than 62 dBA.

Temperature range

Storage temperature range

Storage: -10°C to +50°C (14°F to 122°F).

3.2.2.1 Environmental protection: IP and NEMA ratings

Stainless steel version, medium duty: IP65, NEMA 12.

3.3 Electrical Power



DANGER

Disconnect all power from the metal detector before you work on any of the electrical components of the metal detector. Use properly qualified and trained personnel for all electrical work on the metal detector.

3.3.1 Power circuit protectors and isolators



WARNING

To meet UL requirements, the power circuit protectors and isolator must be UL listed.

3.3.1.1 Power circuit protectors

Use a double pole isolation switch to protect the power supply. Fuse the double pole isolation switch in one of the following ways:

- Double pole circuit breaker, current rating 3 Amp or 4 Amp, type C, 240 VAC.
- Time-lag fuse, current rating 4 Amp or 5 Amp, 250 VAC.

The metal detector has no internal user-replaceable fuses.

There is an internal auxiliary circuit breaker inside the PSU box, for local use only.

3.3.1.2 Isolators

- Install the isolator close to the metal detector and within easy reach of the operator.
- Mark the isolator as the disconnecting device.

3.3.2 Power supply unit



DANGER

Disconnect power before you remove the cover of the CPU IO box.

The power supply unit is in the CPU IO box.

The power supply unit is supplied by mains electricity. It provides DC power to the metal detector, which can be up to three meters away from the CPU IO box.

You can install the power supply unit up to three meters away from the metal detector.

Connect the mains lead as shown:

3.4 Electrical Connections

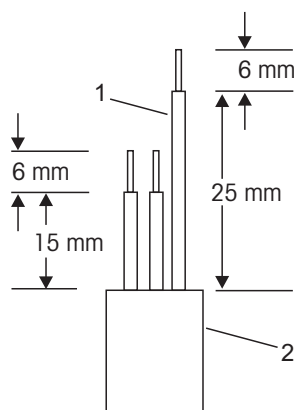
3.4.1 Principal connections

3.4.1.1 Power lead minimum requirements

This section gives the minimum requirements for the power lead. The power lead connects the power to the power supply unit's conclamp.

Item	Details
Power lead cross-section area	0.75 mm ² / 18 AWG

Item	Details
Current rating	6 Ampere
Live wire colour	Brown
Neutral wire colour	Blue
Protective earth conductor wire colour	Green/yellow



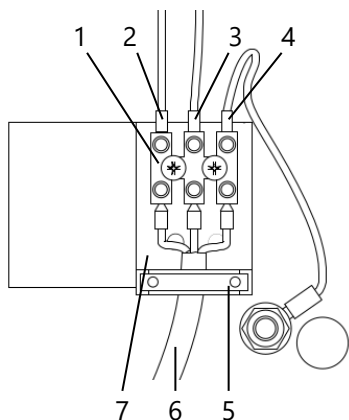
Recommended wire lengths

1	Protective earth conductor wire	2	Power lead sheath
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3.4.1.2 Power lead wiring

This section explains how to connect the power lead to the power supply unit in the CPU IO box.

Wiring power supply unit



Wiring for power supply unit

1	Terminal block	5	Sheath clamp
2	Brown: Live wire (L)	6	Incoming power cable
3	Blue: Neutral wire (N)	7	Conclamp assembly
4	Green/yellow: Protective earth conductor wire		



NOTICE

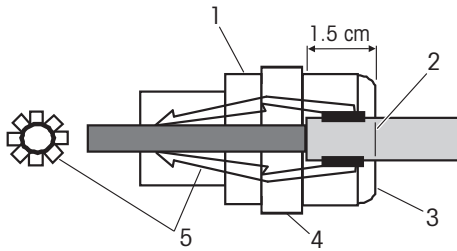
The recommended torque settings for the screw terminals are 0.8 Nm (7 lb in).

3.4.1.3 Metal detector cable wiring

Cable gland assembly

Here is an example of a correct cable gland assembly. The cable gland assembly attaches the metal detector cable to the power supply unit box.

Before you assemble the cable gland, make a location mark on the cable 1.5cm from the cable screen.



Cable gland assembly

1	Main body of gland	2	Location mark on cable, lined up with outer face of nut
3	Nut (torque = 5.5 Nm)	4	Spacer
5	Metallised prongs providing screen termination		



NOTICE

The metallised prongs require an adequate connection to the cable screen for correct metal detector operation.

3.4.2 CPU and User I/O board



CAUTION

Take care to avoid inductive overloads when connecting to external circuits.

When you connect inductive loads that are intended to be driven via the relay switching contacts, make sure that adequate transient suppression is incorporated into external circuit designs.

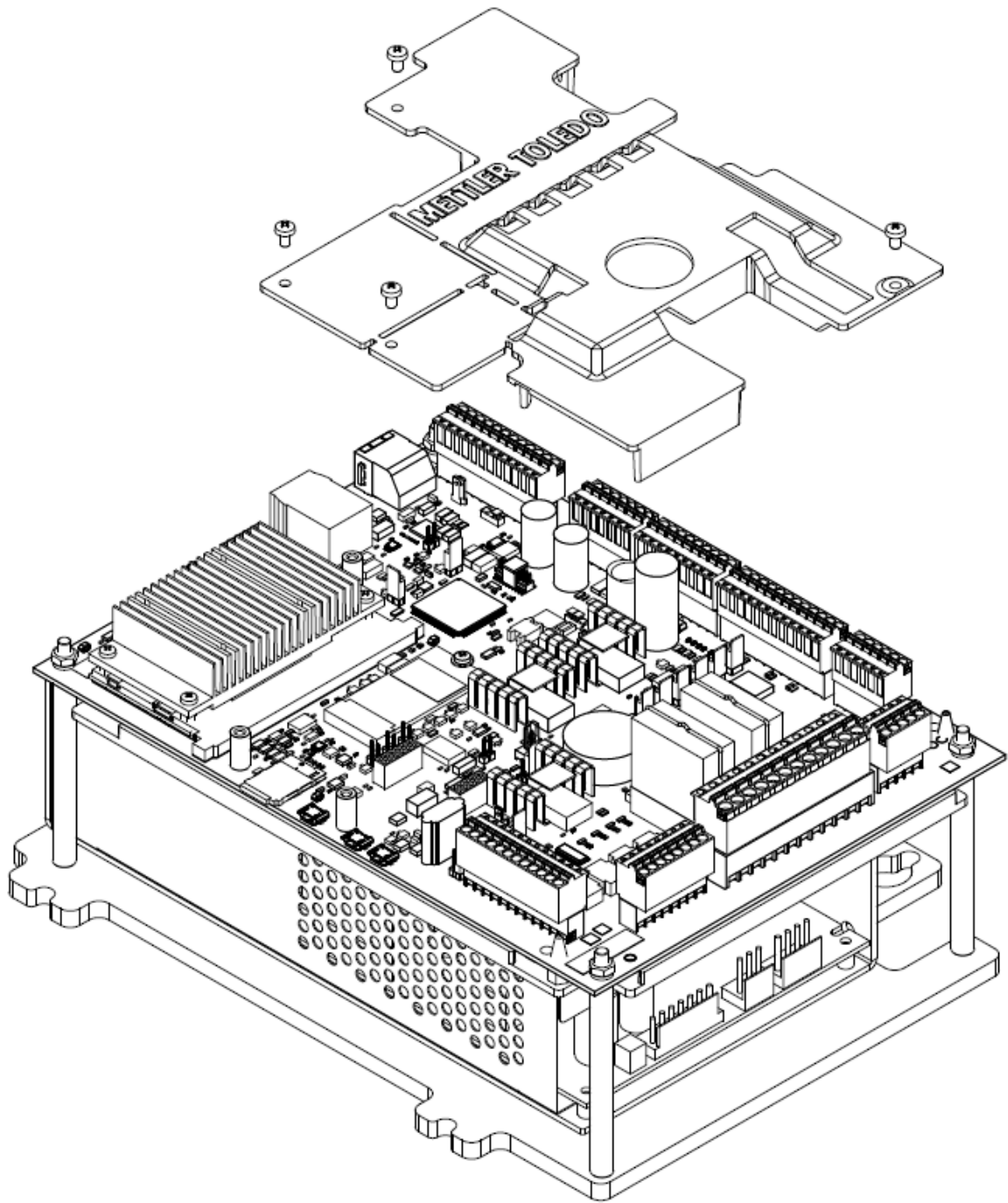
This applies to the solid state relay outputs 1, 2, 3 and 4 and the electromechanical relays of the programmable outputs 5 and 6. Failure to implement suitable back EMF suppression can damage the equipment. Residual transient voltages that are generated from the inductive loads can cause arcing of the relay switching contacts, premature aging of equipment, logic failures and damage or failure of components.

It is not possible for METTLER TOLEDO to make recommendations for all the possible output applications. There are several methods of transient suppression available including transient voltage suppressors, protection diodes and snubber networks that should be used as appropriate.

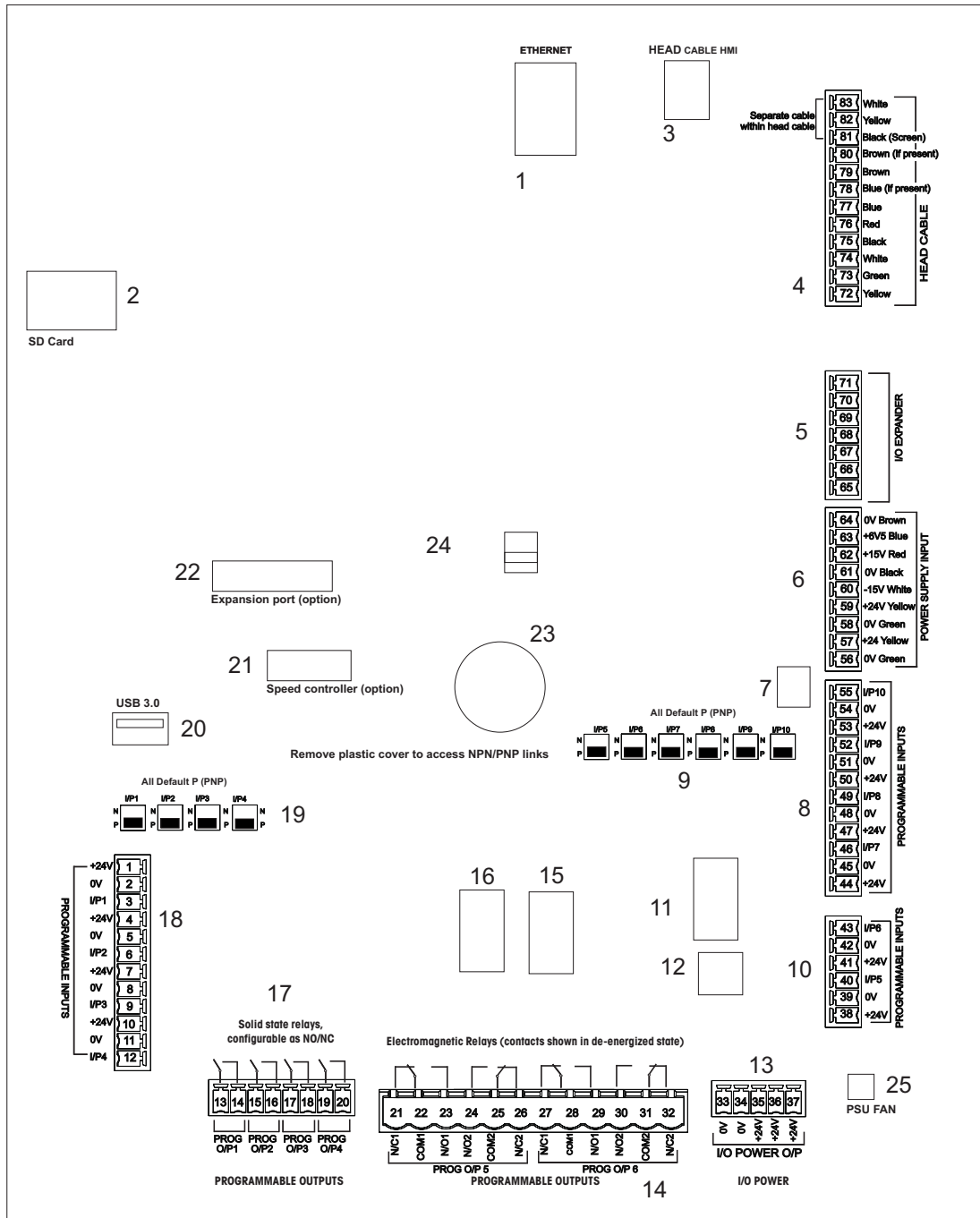
The CPU IO board is in the power supply unit (see item 6 on this typical installation: [Typical installation ► Page 16]).

To access the CPU IO board:

- 1 Open the cover of the box of the CPU IO board and power supply.
- 2 Remove the four securing bolts from the cover on the CPU IO board, as shown:



The following figure shows the CPU IO board with programmable user input/output connections and other connections.



Key	Description	Key	Description
1	Ethernet connection RJ-45	13	I/O power supply outputs
2	SD card	14	Programmable outputs 5 and 6
3	HMI connection RJ-45	15	Electro-mechanical relay for Programmable Output 6
4	Head cable connection	16	Electro-mechanical relay for Programmable Output 5
5	Connection to I/O expander	17	Programmable outputs 1 to 4
6	Power supply input	18	Programmable inputs 1 to 4

Key	Description	Key	Description
7	Power supply status indicators	19	Select NPN/PNP for inputs 1 to 4
8	Programable inputs 7 to 10	20	USB
9	Select NPN/PNP for inputs 5 to 10	21	Speed controller (option)
10	Programable inputs 5 and 6	22	Expansion port (option)
11	Input status indicators	23	Battery
12	Output status indicators	24	Battery link
		25	PSU fan

Notes:

- All inputs are configurable as NPN or PNP. The default state is all inputs set to PNP.
 - Programable outputs are solid state, except 5 and 6 (see item 15 and 16 in the table above).
 - Configure the inputs and outputs in the software through the user interface: [System Integration - Inputs and Outputs ► Page 27]
- 1 Put the cover back on the CPU IO board, and secure with the four screws.
 - 2 Close the cover of the box of the CPU IO board and power supply. **Torque the bolts to 5 N/m.**

3.5 Fitting the Metal Detector to a System

This section contains information on fitting the metal detector to a system.

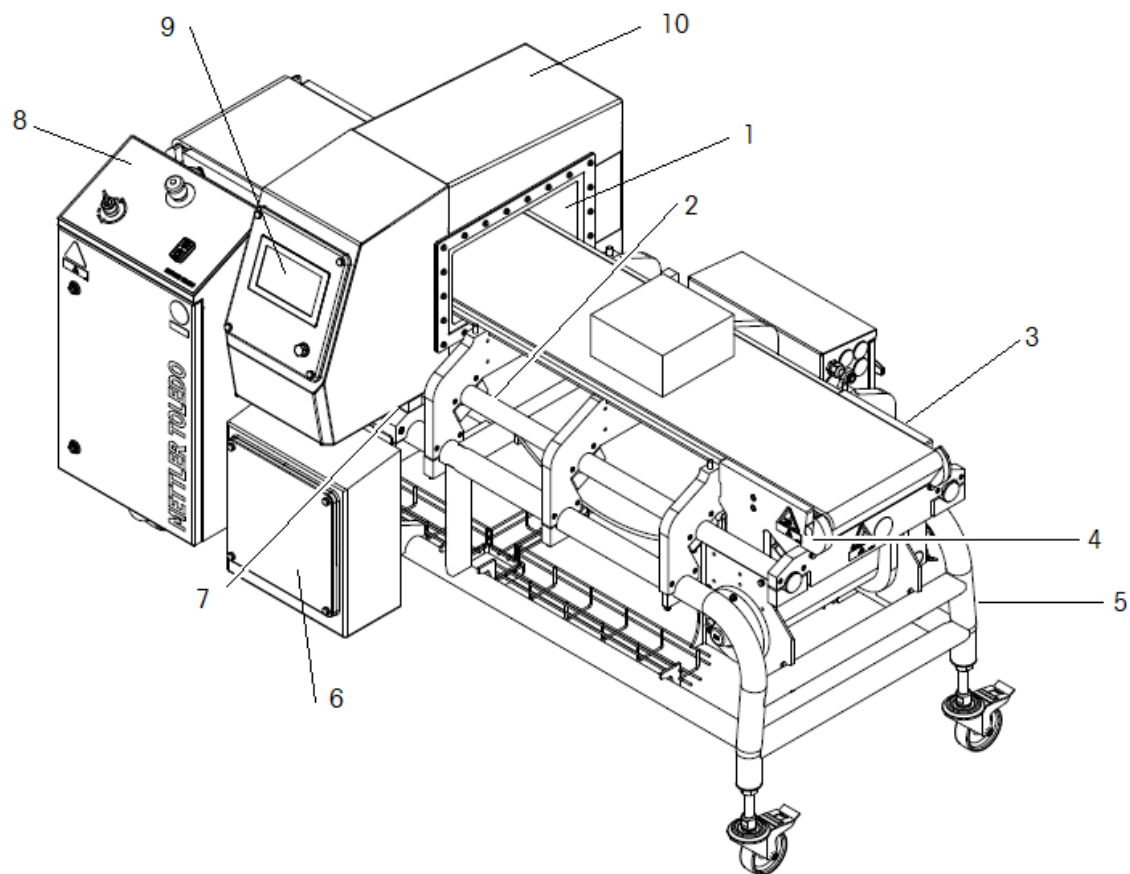
If the metal detector is part of a METTLER TOLEDO system, read the User Manual for the system.

3.5.1 Support framework

3.5.1.1 Typical installation

Follow these guidelines for best performance:

- Make sure the frame support bar (2, in the following figure) are the maximum possible distance from the detector's aperture (1).
- Insulate one end of the roller (4) from the support framework (5) to avoid earth loops. Ground (earth) the other end of the roller. Grounds (earth) all rollers on the same side. We show two insulation examples in the figure 'Bearing block and mounting foot insulation' which follows the figure 'Typical installation' below.
- For the support framework (5), use a welded construction. Do not use bolted sections.
- Insulate all conductive items bolted to the frame using non-conducting insulators. This applies particularly to items positioned across the conveyor such as guards. Always make a ground (earth) connection. Use only one ground connection, to avoid ground loops.
- Install the metal detector (10) onto its support framework (5) using the four plastic mounting feet (7) supplied with the metal detector.
- Earth the metal detector (10) through the power supply unit's protective earth conductor only. Do not add a separate earth to the metal detector.



Typical installation

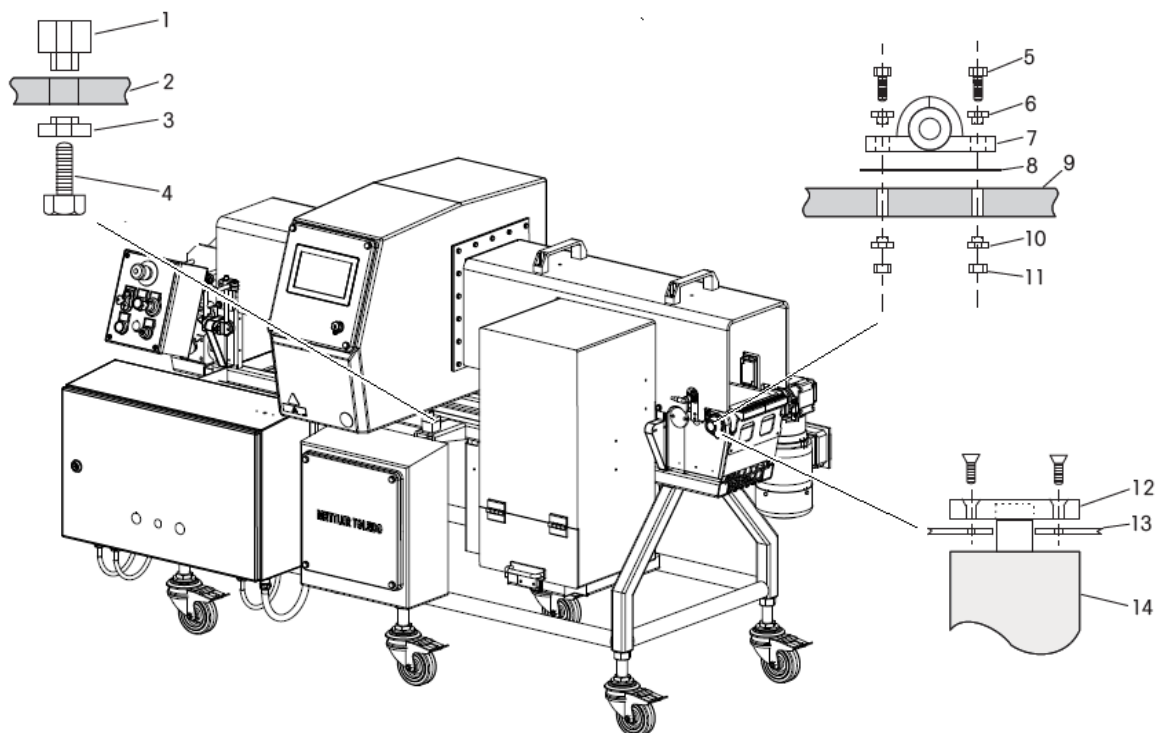
1	Aperture	6	Power supply unit
2	Support bar	7	Plastic mounting feet
3	Non-metallic skidplate	8	Conveyor belt control unit
4	Insulated bearings	9	User Interface
5	Welded support framework	10	Metal detector



NOTICE

Do not electrically bond the metal detector (10) to the support structure. The metal detector (10) is grounded (earthed) only through its own electrical connections to the power supply unit. Do not add a separate earth (ground).

3.5.1.1.1 Bearing block and mounting foot insulation



Bearing block and mounting foot insulation

Mounting foot isolation		Isolate the roller, example 2: Plastic insulation plate and washers on the bearing mounting block	
1	Plastic mounting foot	5	Metal bolt
2	Metal mounting bar	6	Plastic top hat
3	Plastic top hat	7	Metal bearing block
4	Metal bolt	8	Plastic shim
Isolate the roller, example 1: plastic extension to the end of the roller shaft		9	Metal support
12	Plastic roller support	10	Plastic top hat
13	Metal support	11	Metal nut
14	Roller		

4 Operation

This section gives you information for the basic operation of the metal detector. See the **Operation Manual** for details. In this section you will find the default users names and passwords, and enough information about the home screens and the main screens to get started.

4.1 System Integration

M30 R-series of metal detectors provide increased levels of conveyor integration. Central software control replaces the mechanical arrangements and dependencies that were necessary on previous series of metal detectors. There are three combinations possible for conveyor controls:

1. Software control only through the User Interface (UI).
2. Software control through the UI, and Control Panel.
3. Control Panel only.

From the **System Integration** menu, you can select the machine state and control regime required.

If an M30 R-series detector is purchased with a METTLER TOLEDO system, it is fully integrated with System IO and defaults to the best-practice alarm mechanisms. The metal detector monitors all aspects of the system including sensors, conveyor status and safety circuits and generates informative alarms that, where appropriate, interact with the machine mode & state.

Input and output signals monitor the operating environment and make sure that the metal detector does not operate if the conveyor system is stopped or faulty.

The software of the metal detector monitors the inputs and outputs the conveyor system and sensors. The software can automatically start and stop the conveyor system, and monitors a hierarchy of **Alarms**. If a critical fault, or a fault with a high alarm status develops, the software will stop the conveyor system.

Physical stop/start buttons produce logical states that are directly inputted to the system software.

The electrical inverter that drives the conveyor motor only activates following the calculations of the software, according to its configuration.

M30 R-series metal detectors have an expandable I/O system to support the software centric approach.



WARNING

Inhibited functionality

When the **System Integration** menu is open, the 12 **Reject** functionality and some of the alarms are inhibited.

See also

- System Integration - Inputs and Outputs ► Page 27
- Conveyor Control Options (Start/Stop Control) ► Page 27

4.2 Description of the User Interface (UI)

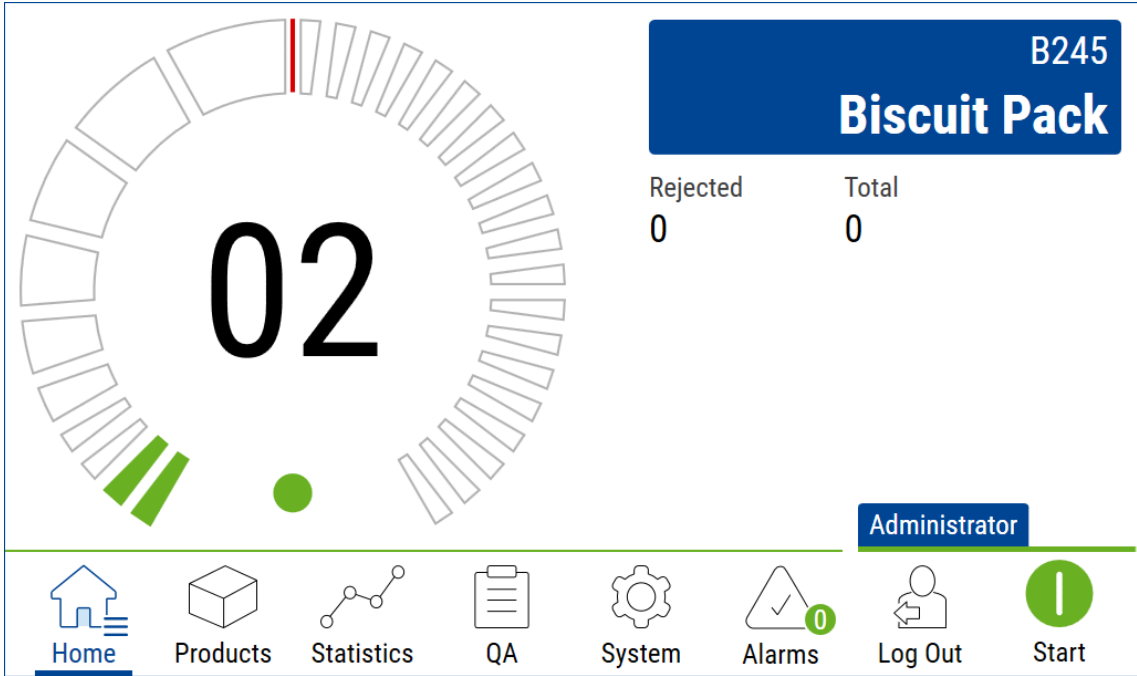
Before you log in, you have access to the **Home** screen and the other main information screens. The user interface (UI) is a touch screen. To operate any element on the screen, touch it once with your finger.



NOTICE

Use your fingers, with or without gloves, to press the keys on the touch screen. Do not use sharp objects.

The main parts of the screen are as follows:



Key	Explanation
 Bargraph	The product's signal strength. The bars past the 12 o'clock position are red, and indicate a detection.
	The Active Product. You can press the icon to go directly to the Changeover menu.
 Main Navigator	The navigation bar. Each icon links to a menu.
 Start Conveyor Start icon  Stop Conveyor Stop icon	In the System settings, you can configure how the software interacts with the conveyor. In the Conveyor menu you can choose to use the Start/Stop Control of the Conveyor on the UI (User Interface). If you do this an additional icon appears to the right of the Log In icon. Start appears if the conveyor is stationary. Stop appears if the conveyor is running. If the button is gray, or if the button is not present, you cannot start or stop the conveyor on the UI. The button becomes gray if the software determines there are underlying conditions or alarms that mean that you should not start the conveyor.

General condition of the metal detector

The **Alarms** icon informs you of the present running condition of the metal detector. The five possible states are:

	System Healthy.
	The metal detector is working, but you are notified that there is a condition to be aware of. The number in the red circle indicates the number of notifications. Pressing the icon will list the notifications.
	The metal detector needs attention. A fault could develop. The number in the red circle indicates the number of notifications. Pressing the icon will list the notifications.
	The metal detector has stopped working. The number in the red circle indicates the number of notifications. Pressing the icon will list the notifications.
	The metal detector has developed a Critical fault. This appears for faults related to safety issues, such as Safety Device Operated .

The menus and pages

When you click on a icon on the navigator, it opens a menu of features. Some of these features have several pages behind them.

1. Previous Page. Press to go back to page 1 of 3 (in the example shown above)	Page (2 of 3) Current page	Next Page Press to go forward to page 3 of 3 (in example shown above)
--	--------------------------------------	---

The Collapsed-Menu Symbol:

The collapsed-menu symbol has three small horizontal lines. The symbol indicates that there are further menus of screens available by clicking on the symbol.



1. Example of the collapsed-menu symbol, on top of the **Home** button.

To go to the next page, use the forward arrow: **»**. To go to the previous page, use the back arrow: **«**

Breadcrumb Navigation

The User Interface used the 'breadcrumb trail' convention. This reminds you how you navigated to the current screen. For instance, in the example below, you are on the **Language** screen, from which you arrived via **Region Settings**.

« System
Region Settings / Language

4.3 The Home Screens

The home screens are screens that all users can view, whether or not they are logged in.

To see the home screens, press the **Home** button. If the triple-bar symbol is visible, press it to see the **Home Menu**.



1. Triple-bar symbol.

4.4 Logging into the User Menus

1 Press the **Log In** icon.

2 Select your **Username**.

➔ NOTE: Your role is determined from your account **Username**.

3 Type in your **Password**.

NOTES:

- The passwords are set up by the **Administrator**, or you can set your own password when you are logged in: [All Default Usernames and Passwords ► Page 22]
- When you log in, the **Log In** icon changes to a **Log Out** icon, and your **Username** appears above the **Log Out** icon.

4.5 All Default Usernames and Passwords

Our StandardLine metal detectors come with up to 10 configurable user accounts. PlusLine metal detectors can have up to 50 configurable accounts.

The table below shows the default passwords for the five pre-configured user accounts (one for each user role).

Username	Operator	Supervisor	Engineer	QA	Administrator
Default password	1111	2222	3333	5555	9999

Administrator can change default pass codes. Contact your local **METTLER TOLEDO Safeline** service representative if you change and lose the **Administrator** password.

All users can change their own password. Go to **System / Users / My Account**.

For further information see Operator manual Managing Accounts and Users.

4.6 Managing Products

4.6.1 Setting up products

Metal Detector	Access Level Required
All	Engineer or Administrator

To create a new product, you can use the setup wizard, or copy an existing product and change its name.

To access the setup wizard, go to **Products > Product List**, then either select **New**, or open an existing product and select **Setup Wizard**.

If you are setting up a product for the first time, simply go to the **Products** page and select **Setup Wizard**.

Notes:

- You must set up the **Speed Settings** before you can set up a product. See Setting up the Speed Settings and Speed Sensor. Note also that if you adjust **Speed Settings**, it will invalidate the settings for existing **Products**. Therefore, **Speed Settings** should only be adjusted on initial setup or for a major reconfiguration of the metal detection system.

- The recommended way to set up a product is to pass it through the aperture of the metal detector, and use the **Auto Setup** feature to calculate the best **Sensitivity**, **Phase** and **Mode** for the product.
- During the **Auto Setup**, the reject mechanism is usually disabled to make sure that no products are unnecessarily rejected during a product setup.
- If you are working in a live production environment, you can disable **Reject Inhibit** (in **System/Auto Setup Settings**) to make sure that contaminated product is rejected from the production line. However, this is only possible if you are confident that the active product settings are valid, and correctly set up.

Auto Setup gives you instructions on the screen to pass products. The product must be moving (not stationary). If the products are discrete products, they must be spaced.

If setting up M31R, M33R or M33RB, each step of **Auto Setup** can also be done as a manual process in the product settings of active products. If setting up M34R, you must run **Auto Setup** at least once for each product.

Prerequisites:

- You need to have some of the product you want to set up, in the form and quantity that will normally pass through the aperture.
- You need an approved metal test sample.
- If a pack sensor is fitted, it must be correctly set up.
- The metal detector speed, and the speed sensor (if used), must be correctly set up.
- You must be logged in as **Administrator** or **Engineer**.

- 1 Press **Products**.
- 2 Press **Product List**.
- 3 Enter the **Name** of the product for a new product. You can edit the name of an existing product.
- 4 Press **New** for a new product, or press an existing product to launch its **Setup Wizard**.
- 5 Enter the product **Code**.
 - ➔ Note: The product **Code** is usually optional. However, in Product / Product Rules the **Engineer** or the **Administrator** can make the Code mandatory by selecting **Mandatory Product Code Enabled**.
- 6 Press **Next**.
 - ➔ **Result:** If the product is new, the message **Do the setup now?** appears, with the buttons **Save and Exit** and **Continue**.
 - ➔ **Note:** If you create a new product, but do not complete the setup procedure, it will appear in the product list with the text **Requires Setup**:
- 7 Press **Continue**.
 - ➔ Result:
The **Start Auto Setup** screen appears, with a **Mode Selection** field. The default value is usually **Auto**, however, if the speed settings of the system are too slow, **Auto** is not available.
- 8 In the **Mode Selection** menu, select **Auto** to allow the system to calculate the best **Mode**, or manually select a **Mode** from the list.
 - ➔ The **Start** conveyor button will change to green.
- 9 Press the **Start** conveyor button:



10 Press the **Start** button to **Start Auto Setup**:



- 11 Follow all the instructions on the screen until **Auto Setup** is complete. Make sure that the product that you pass through the metal detector is representative of your product range.
- ➔ The product must be passed several times. The metal detector will calculate the optimum **Mode**, **Phase** and **Sensitivity** for wet and dry products.
- 12 After you have passed enough product, press **Done**.
- ➔ The screen **Press Next or Repeat if necessary** appears.
- 13 Press **Next**, or press **Repeat** if you want to repeat the **Auto Setup**.
- ➔ The screen **Adjust inspection settings if necessary** appears. You can manually adjust **Mode**, **Sensitivity** and **Phase**. Press **Advanced Settings** if you want to adjust **Sensitivity 2** and **Phase Tolerance**.
- 14 Press **Next**.
- ➔ The screen appears asking you to set up **Synchronized Reject**. The options depend on the type of **Reject Mechanism** on the conveyor system (set in the **Function/Reject** in **Outputs**) .
 - ➔ **NOTE:** If there is no pack sensor, the **Synchronized Reject** option does not appear.
- 15 Pass samples, according to the instructions on the screen. (The instructions will differ, depending on whether or not you are using **Synchronized Reject**.)
- 16 Press **Next**.
- 17 Pass product with an approved metal sample to make sure it rejects reliably. Adjust the **Detection to Alarm** and **Maximum Pack Length** values if necessary.
- 18 Press **Next**.
- 19 Press **Save**.
- ➔ **Result:** When you press the product **Cancel** and **Changeover** appear at the bottom. To activate the product, press **Changeover**.



NOTICE

If critical metal detector settings are changed, products might require a re-setup

Products settings for existing products might need to be reset if critical system settings change, for example as if a speed sensor is fitted or replaced, or pack sensor for synchronized reject products.

Related Topics:

Setting up the Speed Settings and Speed Sensor

[System Integration - Inputs and Outputs ► Page 27]

4.6.2 Viewing and Changing Products

Metal Detector	Access Level Required
All	Operator, Supervisor, Engineer or Administrator

You can search for a product by pressing the filter icon:  to display a **Search** field ('fuzzy' text search) and a **Sort By** field.

- 1 To change the active product, press **Products**.
- 2 Press the **Changeover** button.

- ➔ **Note:** If you have a higher access level than **Operator**, you can also select **Changeover** by searching the **Product List** and then selecting a product.
- 3 Press to select the new active product from the list.
 - ➔ **Results:**
 - ➔ The **Name** of the running product is displayed in the large icon on the top-right of the screen.
 - ➔ In the Product List, the word **Active** appears on the product icon of the current product.

4.6.3 Editing Products and Product Settings

Metal Detector	Access Level Required
All	Engineer or Administrator (Supervisor has some limited access to product settings.)

Auto Setup and initial adjustments

METTLER TOLEDO recommends that you set up product using the **Auto Setup** feature. However, following **Auto Setup** you may still observe high levels of variation in the product signal on the bar graph resulting in false detections and rejections. If this happens, there are a number of options available to make improvements. You can:

- Run **Auto Setup** again with an increased number of products and more varied presentation.
- Decrease the target bar graph level. (See [Set Signal Level for Auto Setup ► Page 25] below.) Similarly, if the product signal is very consistent, the target bar graph can be increased to improve detection capability.
- Manually change the **Mode** and re-setup. (See [Set Mode Selection manually ► Page 25] below.)
- Manually adjust the product settings. (See [Adjust Sensitivity and Phase manually ► Page 25] below.)

Set Signal Level for Auto Setup

Note: The below procedure requires Administrator level access.

To adjust a product's target bar graph level, open the **Auto Setup Settings** menu and enter a new value in **Signal Level for Auto Setup**.

Set Mode Selection manually

In **Product Settings/Auto Setup**, change the **Auto** option in **Mode Selection** to a different mode (normally a lower mode than it automatically selected) and re-start **Auto Setup**.

Adjust Sensitivity and Phase manually

Open **Inspection Settings** for the product and manually adjust **Sensitivity** and **Phase** to fine-tune the product signal level whilst passing the product. Once the desired bar graph level is reached, accept the changes.



Important Safety Message

Avoid changing Advanced Settings.

To be able to edit **Advanced Settings** successfully, you need special training.

Changes to existing products

Once a product has been set up, you can edit the product **Name**, **Code**, **Inspection Settings**, and **Region Settings** such as **Mode Selection**, **Sensitivity** and **Phase**.

- 1 Go to **Products**. You can edit the **Active** product or select a product from the **Product List**.
- 2 Press **Settings**.

4.6.4 Non-Synchronized Reject Settings

Metal Detector	Access Level Required
All	Engineer or Administrator

The **Reject Settings** for each product is in the **Product Settings** menu.

- **Note:** The product must be active before you can edit **Reject Settings** and other **Product Settings**.

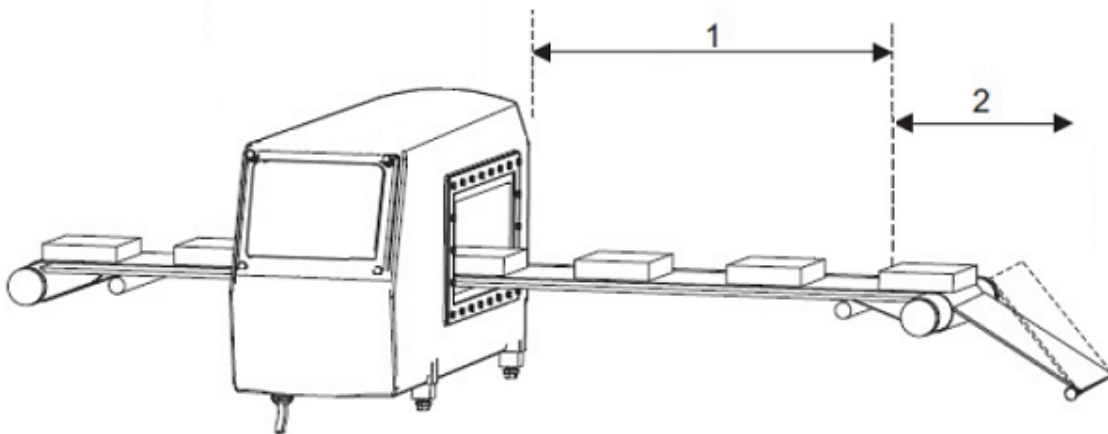
1 Go to **Products / Product Settings / Reject Settings**.

- ➔ **NOTE:** The **Reject Settings** set the reject timings. The timing unit of measurement can be **ms** or, if the metal detector has a Speed Sensor, the unit of measurement can be **pulses**. The units are also determined by a combination of speed sensor and rejector mechanism. The unit of measurement is set at **Administrator** level.

2 Enter the value in the **Detection to Rejection** field (in **ms** or **pulses**). See the figure below.

3 Enter the value in the **Reject On Extension** field (in **ms** or **pulses**). See the figure below.

- ➔ **NOTE:** You can verify this value by passing a metal sample with the product. Adjust the value if necessary.



1	Detection to Rejection	2	Reject On Extension
---	-------------------------------	---	----------------------------

4.6.5 Synchronized Reject Settings

Metal Detector	Access Level Required
All	Engineer or Administrator

Metal detectors that have a **Pack Sensor** can be set to **Synchronized Reject** operation (at the **Administrator** or **Engineer** level). To use this option, press **Synchronized Reject** and follow the instructions on the screen.

- **Prerequisite:** To set up the **Synchronized Reject** you need a pack of your active product and an approved metal test sample that is large enough to consistently trigger a reliable detection. You can only edit the **Active** product.

1 In **Products** go to **Product Settings**. Go to **Reject Settings**.

- ➔ **NOTE:** The **Reject Settings** set the reject timings. The timing unit of measurement can be **ms** or, if the metal detector has a Speed Sensor, the unit of measurement can be **pulses**. The units are also determined by a combination of speed sensor and rejector mechanism. The unit of measurement is set at

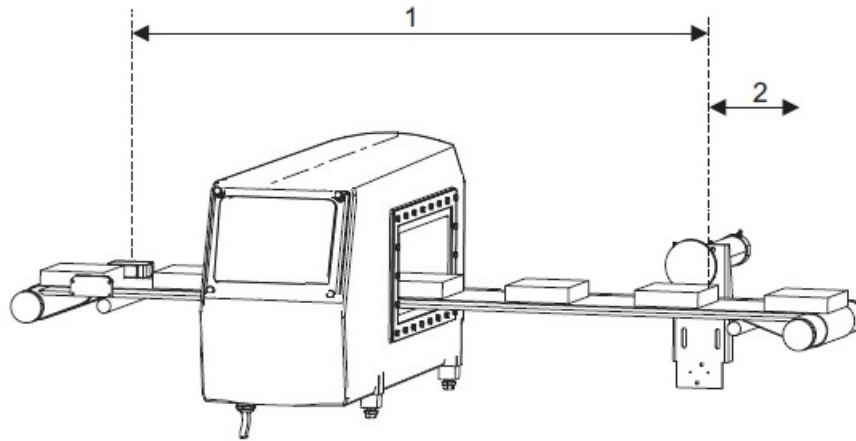
- ➔ **Administrator** level.

2 Press **Set Up Synchronized Reject**, and follow the instructions on the screen.

3 Enter the value in the **Pack to Reject** field (in **ms**). See the figure below.

4 Enter the value in the **Reject On** field (in **ms**). See the figure below.

- ➔ **NOTE:** You can verify this value by passing the product. Adjust the value if necessary.



1	Pack to Reject	2	Reject On
---	----------------	---	-----------

4.7 Conveyor Control Options (Start/Stop Control)

Metal Detector	Access Level Required
All	Administrator

The **Conveyor** screen allows you to choose how you want to start and stop the conveyor. You can choose to have conveyor controls on the User Interface (UI) , or on the conveyor system input/output (IO), or both.

- Precondition: You must be logged in as **Administrator** to access the **System Integration** menu

- 1 Press **System**. Scroll to **System Integration**.
- 2 Press **Conveyor**.
- 3 Select the method or the methods of **Start/Stop Control**. See The Start/Stop Control for further information.
 - **None**. The conveyor cannot be started or stopped from the UI or the software of the metal detector. The PackML state of the metal detector is permanently set to **Execute**.
 - **External**. The conveyor is controlled by an external system, such as a Programmable Logic Controller (PLC). The external system can affect the PackML state of the metal detector.
 - **System IO**. The conveyor is wholly controlled by the software of the metal detector, in response to the programmable Inputs and Outputs.
 - **UI**. The conveyor is started and stopped on the UI.
 - **UI and System IO**. The conveyor is started and stopped on the UI, and in response to the programmable Inputs and Outputs.

Related Topics:

How the Software Works with Programmable Inputs and Outputs

Machine Status (PackML State)

4.8 System Integration - Inputs and Outputs

The **System Integration** screens allow you to set up the input and outputs on the User I/O board. Refer to [CPU and User I/O board ► Page 13] for details of how to access the User I/O board, and where to find the input and outputs where you can wire in external systems and devices.

NOTE: METTLER TOLEDO offers pre-configured settings for commonly-used systems. It is possible that the configuration you need already has the correct **System Integration**.

- Precondition: You must be logged in as **Administrator** to access **System Integration** settings.

- 1 Press **System**. Scroll to **System Integration**.

- 2 Press **Inputs**.
 - ➔ All the
 - ➔ **Inputs** of I/O board appear, with the assigned function or as **Unused**.
- 3 Press on any of the Inputs that is **Unused** or that you want to edit.
- 4 Assign a **Function** from the drop-down list.
- 5 Depending on the **Function** you select, further fields might appear. Put the correct details in these fields.
- 6 Select **Active High** or **Active Low** from the **Active Level** drop-down list.
- 7 Repeat steps 3 to 6 for all the **Inputs** that you want to configure.
- 8 Go back to **System Integration**.
- 9 Press **Outputs**.
 - ➔ All the **Outputs** of I/O board appear, with the assigned function or as **Unused**.
- 10 Press any of the **Outputs** that is **Unused** or that you want to edit.
- 11 Assign a **Function** from the drop-down list.
- 12 Depending on the **Function** you select, further fields might appear. Put the correct details in these fields.
- 13 Repeat steps 10 to 12 for all the **Outputs** that you want to configure.

4.9 Configure the Pack Sensor

The pack sensor (which is usually a light beam) is fitted on the conveyor system near the infeed of the metal detector, but it can be downstream too. For a description of the pack sensor, and how it interacts with the other sensors on the system, refer to your conveyor system manual.

- Preconditions:
 - You must log on as **Administrator**.
- 1 Go to the **System** screen. Scroll to **System Integration**
 - 2 Press **Inputs**.
 - 3 Select the input that you want to use, and that you have wired to the Pack Sensor.
 - 4 Select the **Function** as **Pack Sensor**
 - 5 Select **Active High** or **Active Low**.
 - 6 Enter the distance from the pack sensor to the side of the head, in the **Pack to Head Distance** field.
 - ➔ You will now need to configure the reject settings for each product.
 - 7 Go to the **Products Product List**.
 - 8 Find the product that you want to set up.
 - 9 Enter the correct values in the **Detection to Rejection** and **Reject On Extension** fields.
 - 10 Enter the **Pack Length** and the **Maximum Pack Length**.

5 Routine Maintenance

5.1 Cleaning



NOTICE

Use the following advice as part of your standard operating procedures to avoid damaging the metal detector when cleaning.



WARNING

Only trained and competent personnel are permitted to sanitize the metal detection system.

If the metal detector is part of a **METTLER TOLEDO Safeline** system, follow the cleaning instructions in the System User Manual.

Follow your facility's operating and cleaning procedures when cleaning the metal detector. Consider the following guidelines:

- Make sure all cleaning procedures do not exceed the Ingress Protection (IP) rating of the metal detection system. For your metal detector's IP ratings, see: Environmental protection: IP and NEMA ratings. Cleaning procedures that exceed the metal detection system's IP rating could invalidate the metal detector's warranty and could result in early failure.
- Do not remove stubborn contaminants with a close-proximity water jet.
- Do not use any abrasive or corrosive product that could damage metal detector's surface finish. This may cause the surface to rust or corrode.
- Salt water or sea water can cause the surface to rust or corrode.
- Do not use any solvent-based or chloride-based cleaners. They may cause the surface to rust and the aperture lining to corrode.

5.2 Disposal

In conformance with the European Directive 2012/19/EU on Waste Electrical and Electronic Equipment (WEEE), you must not dispose of any equipment with this label in domestic waste. This also applies to countries outside the EU, according to their specific requirements.

Please dispose of any equipment with this label in accordance with local regulations at the collecting point specified for electrical and electronic equipment. If you have any questions, please contact the responsible authority or the distributor from which you purchased this product. If you pass on this system to other parties (for private or professional use), the content of this regulation must also be relayed.

Thank you for your contribution to environmental protection.



5.3 Metal Detector Information

Every metal detector has a serial number. You will need this number, and other basic information about the metal detector, if you want to contact METTLER TOLEDO Service with a query or to report a fault.

Click on **System**, and then click on **Information**. The **Information** page tells you the **Serial Number**, the **Model Type**, the **Software Version**, and other basic details.

6 Technical Data

6.1 Technology

High frequency, low power electromagnetic coil system, monitored and controlled by a microprocessor system.

6.2 Operating frequencies

Ranging from 10 kHz to 1 MHz.

6.3 Power input

Voltage

100 to 240 VAC (+10% / -15%).

Power supply voltage fluctuations not to exceed +10% or -15% of the stated voltage.

Power

140 W maximum.

Frequency

50 to 60 Hz.

Transient overvoltage category

Transient overvoltage category II up to 300 V.

6.4 Environmental conditions

Humidity range

Maximum relative humidity: 93% for temperatures up to 45°C.

Indoor altitude

Up to 2000 m.

Pollution degree

Pollution degree 2 in accordance with IEC 664.

Warm-up time

No warm-up time required at an ambient temperature of 20°C.

Sound output

Less than 62 dBA at a distance of 1 m, unless you install an audible alarm louder than 62 dBA.

6.5 Environmental protection and IP Rating



NOTICE

To achieve the specified environmental protection, torque the bolts on the control module and the lid of the power supply unit box to 5 Nm (45 in.lbs).

IP Rating: M30 R-Series metal detectors are available in IP rating options of IP65, IP66 and IP69K.

6.6 Temperature range

Operating: -10°C to +45°C (14°F to 113°F).

Storage: -10°C to +50°C (14°F to 122°F).

6.7 Protection and standards

Impact rating

IK08.

Insulation ratings

Insulation ratings of relays on the user input/output board:

- Mains input to earth: Reinforced.
- Contact-to-contact relays: Basic.
- Pole-to-pole relays: Reinforced.
- Remaining user terminals to the earth: Reinforced (all are secondary circuits derived from power circuits of overvoltage category II, up to 300V).

[CPU and User I/O board ► Page 13]

Internal battery

Battery life: Typically 10 years.

Nominal voltage: 3 VDC.

Lithium primary battery on module CPU board. Safe disposal necessary as defined by your local regulations.

See also

 CPU and User I/O board ► Page 13

6.8 Inputs

Programmable Inputs 1 to 10

Operating voltage	5 to 35 VDC
Operating current	N/A
Output type	NPN, PNP or voltage-free contact

6.9 Outputs

Programmable Outputs 1 to 4

Solid state relay.

Maximum Current: 1 A.

Maximum Voltage: 48 VDC or 48 VAC peak.

Programmable Outputs 5 and 6

Volt-free changeover contacts. Operated by the metal detector's programmable reject timer, following the detection of metal.

Maximum power: 500 W.

Maximum current: 3 A (non-inductive).

Maximum voltage: 250 VAC or 30 VDC (non-inductive).

Minimum load requirement for reliable contact switching: 300mW.

6.10 Communication

Use a suitable shielded twisted pair cable with screen for communication connections.

7 Troubleshooting

Noise instability

If the metal detector shows evidence of noise, such as an unsteady signal strength indicator, or false triggers, check for the following:

- Noise from external sources, or ground (earth) loops, or excessive vibration. Refer to the chapter Requirements and Restrictions for Metal Detection Technology.
- Internal cabling and connections. Make sure that the connections and cables are secure within the modules, and between the modules.
- If you are sure that the noise is not caused by external sources or insecure connections, replace the head control module: Replacing the Head Control Module.

User Interface Problems

If there is no display on the user interface, if the touch screen is not working correctly, check for the following:

- Make sure that the screen is calibrated. To do this, press and hold on the screen at the same time as starting up the metal detector. Follow the calibration instructions on the screen.
- Make sure that the two RJ-45 connectors between the display and the interconnection board are secure.
- Make sure that the cable from the CPU IO module to the metal detector is secure.
- If you have done the above checks, and the problem remains, change replace the HMI unit: Replacing the HMI Unit.

Input and Output Problems

- Check the supply voltage.
- Check the wiring.
- Check the configuration in the software: [System Integration - Inputs and Outputs ► Page 27]
- If you have done the above checks, and the problem remains, change the CPU IO module: Replacing the Power Supply Unit and CPU IO Module

See also

 System Integration - Inputs and Outputs ► Page 27

To protect your product's future:

METTLER TOLEDO Service assures the quality, measuring accuracy and preservation of value of this product for years to come.

Please request full details about our attractive terms of service.

www.mt.com/slmd-contact

For more information

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Subject to technical changes.
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